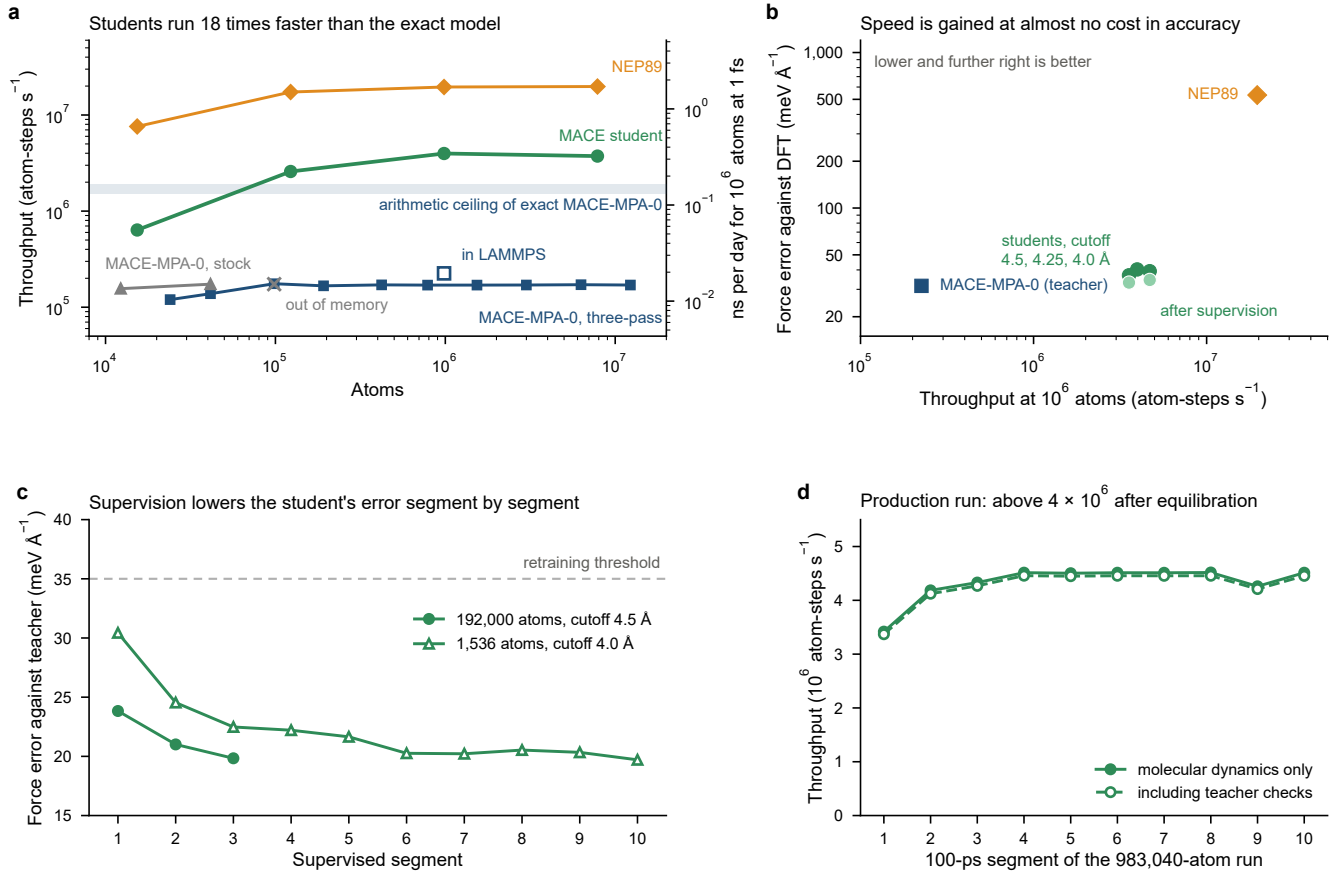
a
Students run 18 times faster than the exact model
Throughput (atom-steps s^{-1})
ns per day for 10^6 atoms at 1 fs
Atoms
NEP89
MACE student
arithmetic ceiling of exact MACE-MPA-0
in LAMMPS
MACE-MPA-0, stock
out of memory
MACE-MPA-0, three-pass
b
Speed is gained at almost no cost in accuracy
lower and further right is better
Force error against DFT (meV Å^{-1})
Throughput at 10^6 atoms (atom-steps s^{-1})
NEP89
students, cutoff
4.5, 4.25, 4.0 Å
MACE-MPA-0 (teacher)
after supervision
c
Supervision lowers the student's error segment by segment
Force error against teacher (meV Å^{-1})
Supervised segment
retraining threshold
192,000 atoms, cutoff 4.5 Å
1,536 atoms, cutoff 4.0 Å
d
Production run: above 4 × 10^6 after equilibration
Throughput (10^6 atom-steps s^{-1})
100-ps segment of the 983,040-atom run
molecular dynamics only
including teacher checks